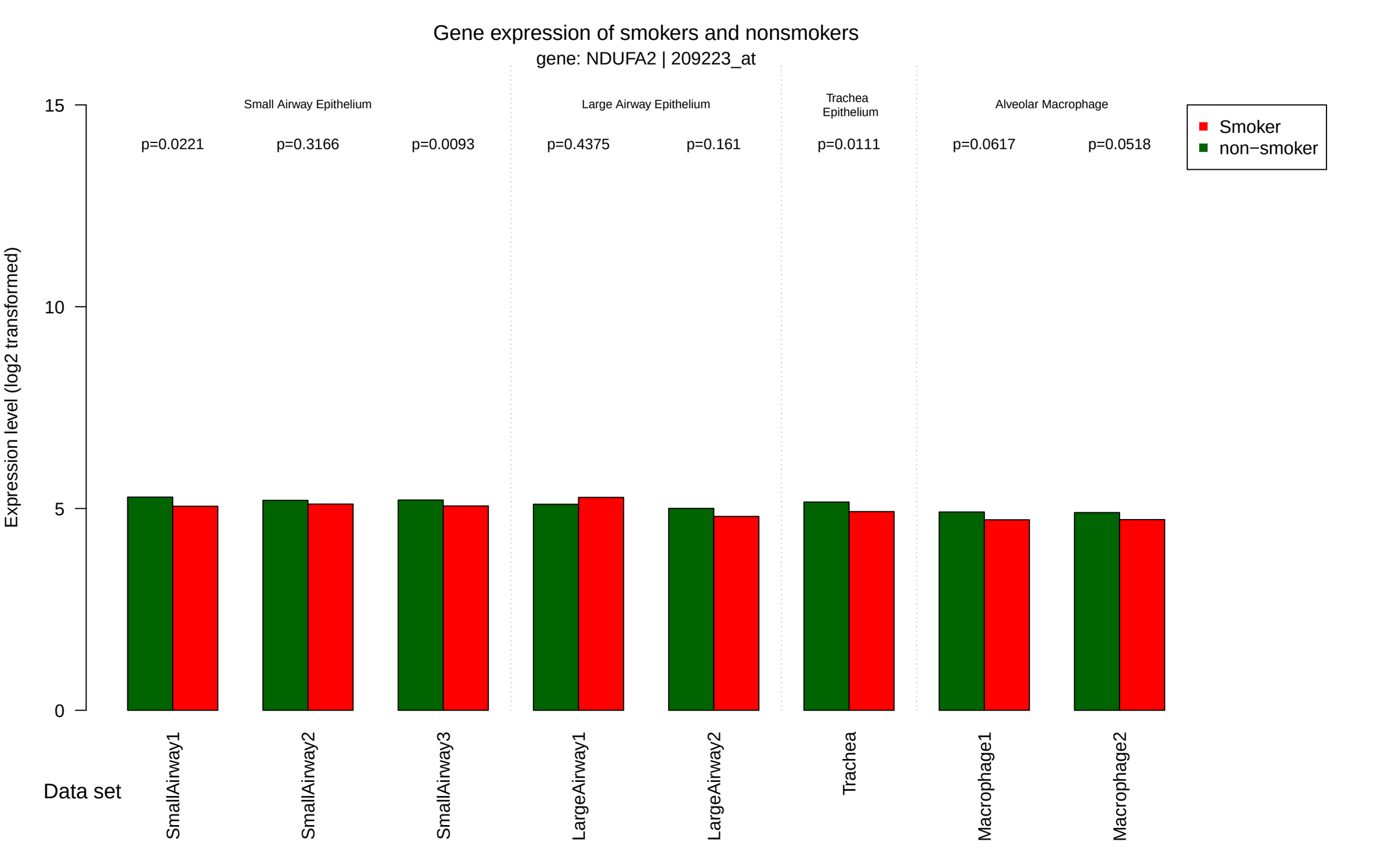
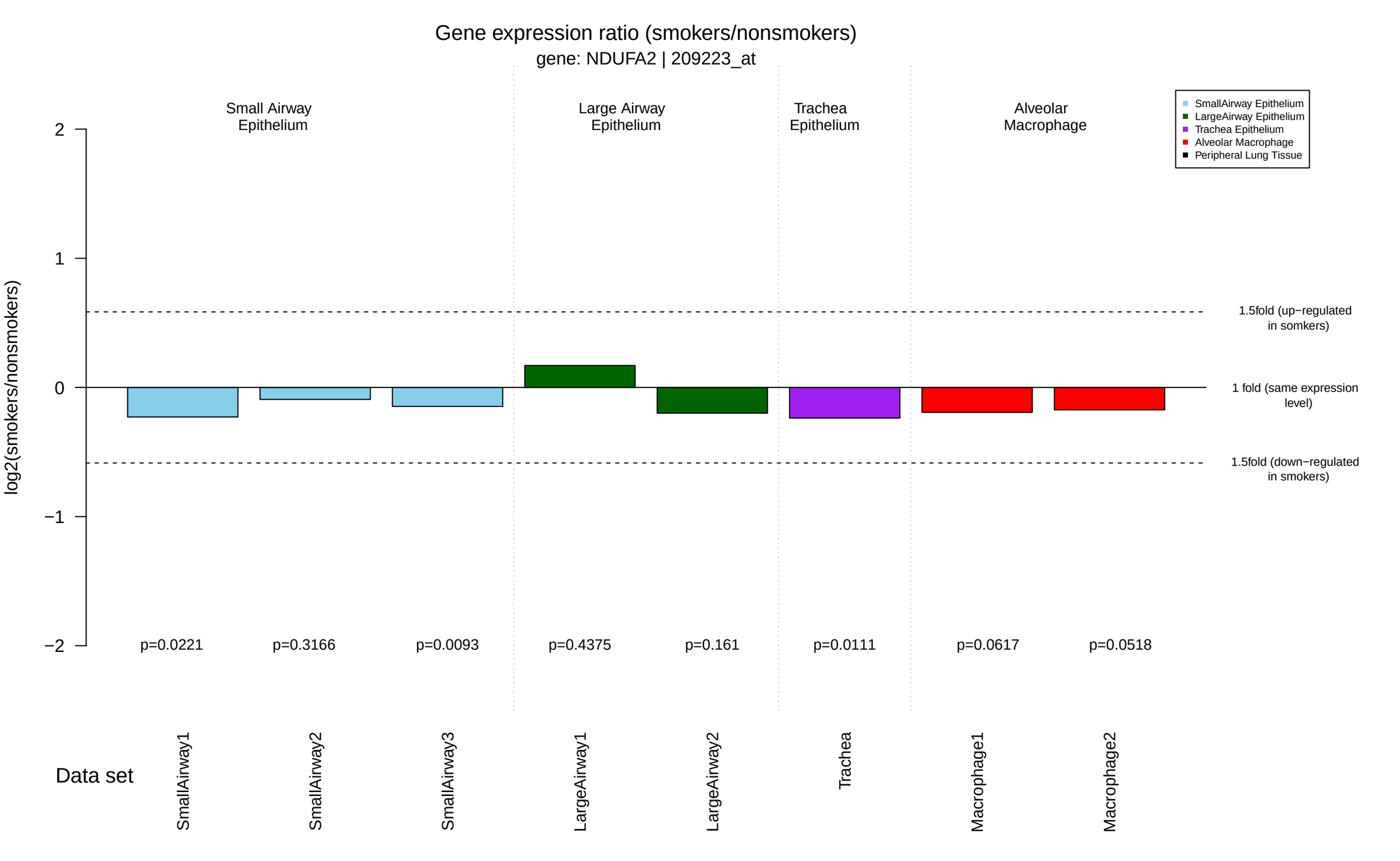
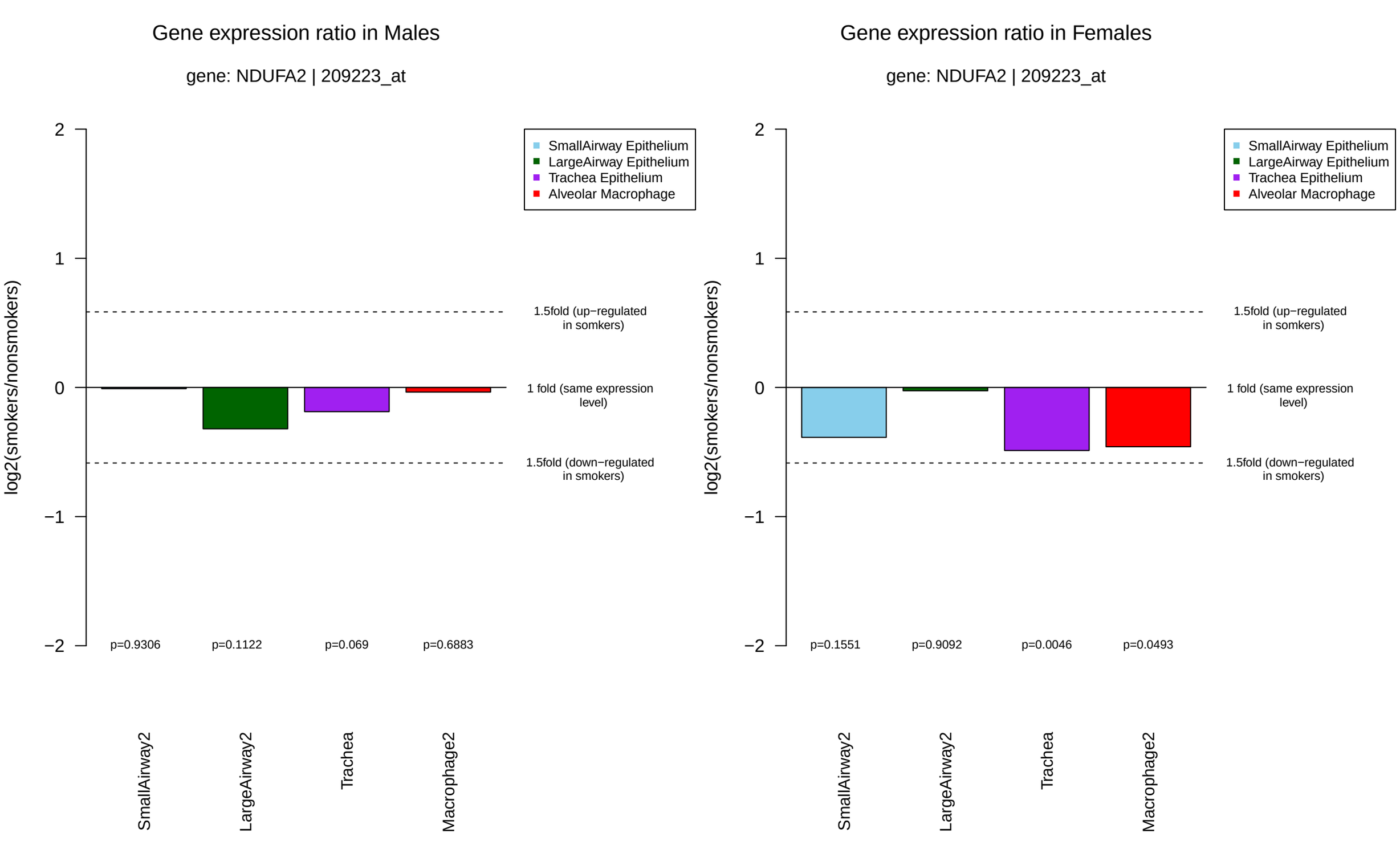
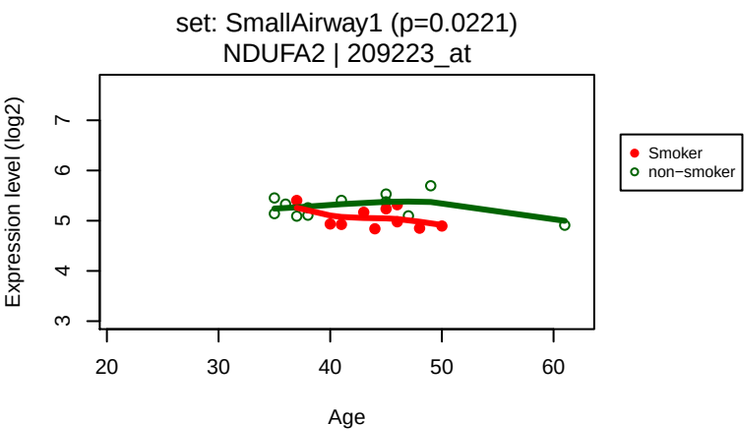
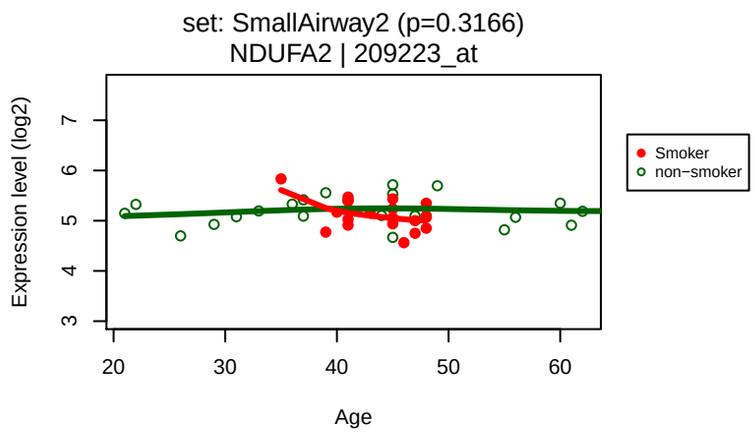
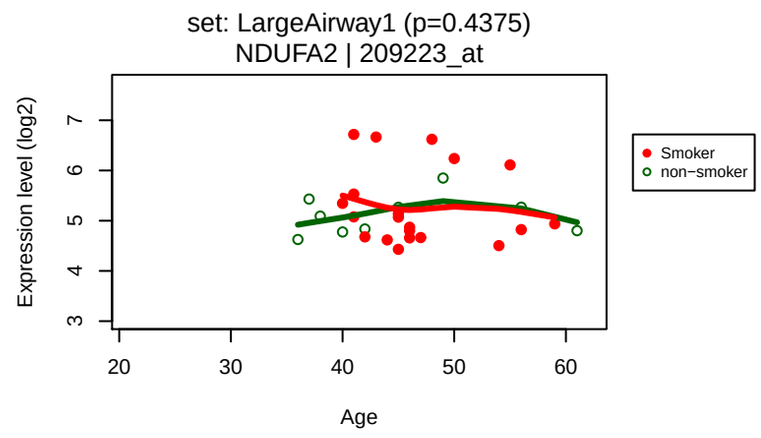

Gene expression of smokers and nonsmokers
gene: NDUFA2 | 209223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0221
p=0.3166
p=0.0093
p=0.4375
p=0.161
p=0.0111
p=0.0617
p=0.0518
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

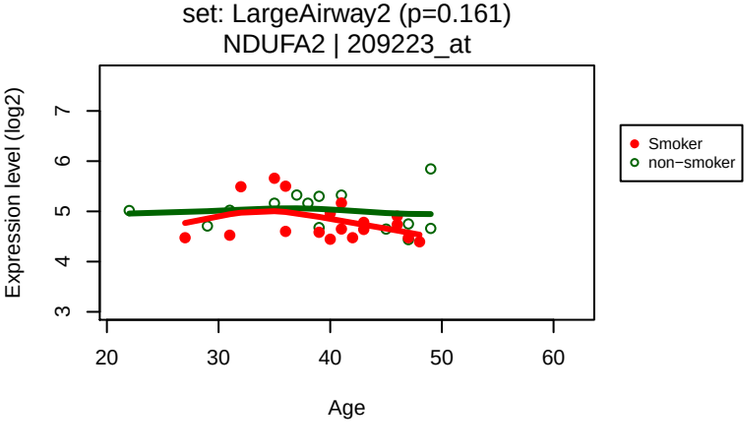

Gene expression ratio (smokers/nonsmokers)
gene: NDUFA2 | 209223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0221
p=0.3166
p=0.0093
p=0.4375
p=0.161
p=0.0111
p=0.0617
p=0.0518
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

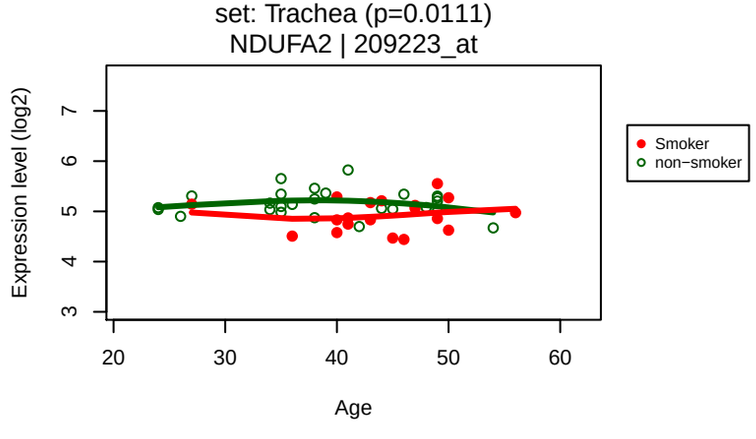

Gene expression ratio in Males
gene: NDUFA2 | 209223_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9306
p=0.1122
p=0.069
p=0.6883
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: NDUFA2 | 209223_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1551
p=0.9092
p=0.0046
p=0.0493
SmallAirway2
LargeAirway2
Trachea
Macrophage2

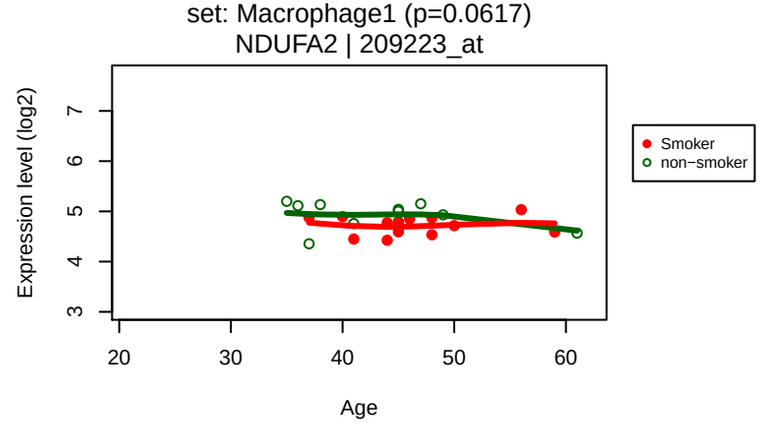

set: SmallAirway1 (p=0.0221)
NDUFA2 | 209223_at
Expression level (log2)
Age
Smoker
non−smoker

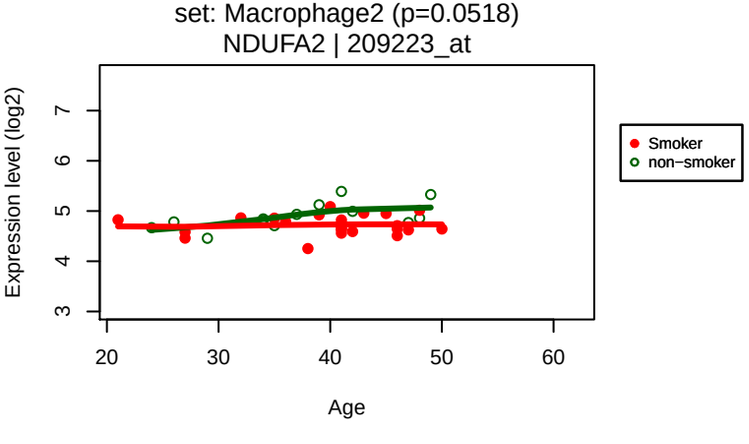

set: SmallAirway2 (p=0.3166)
NDUFA2 | 209223_at
Expression level (log2)
Age
Smoker
non−smoker

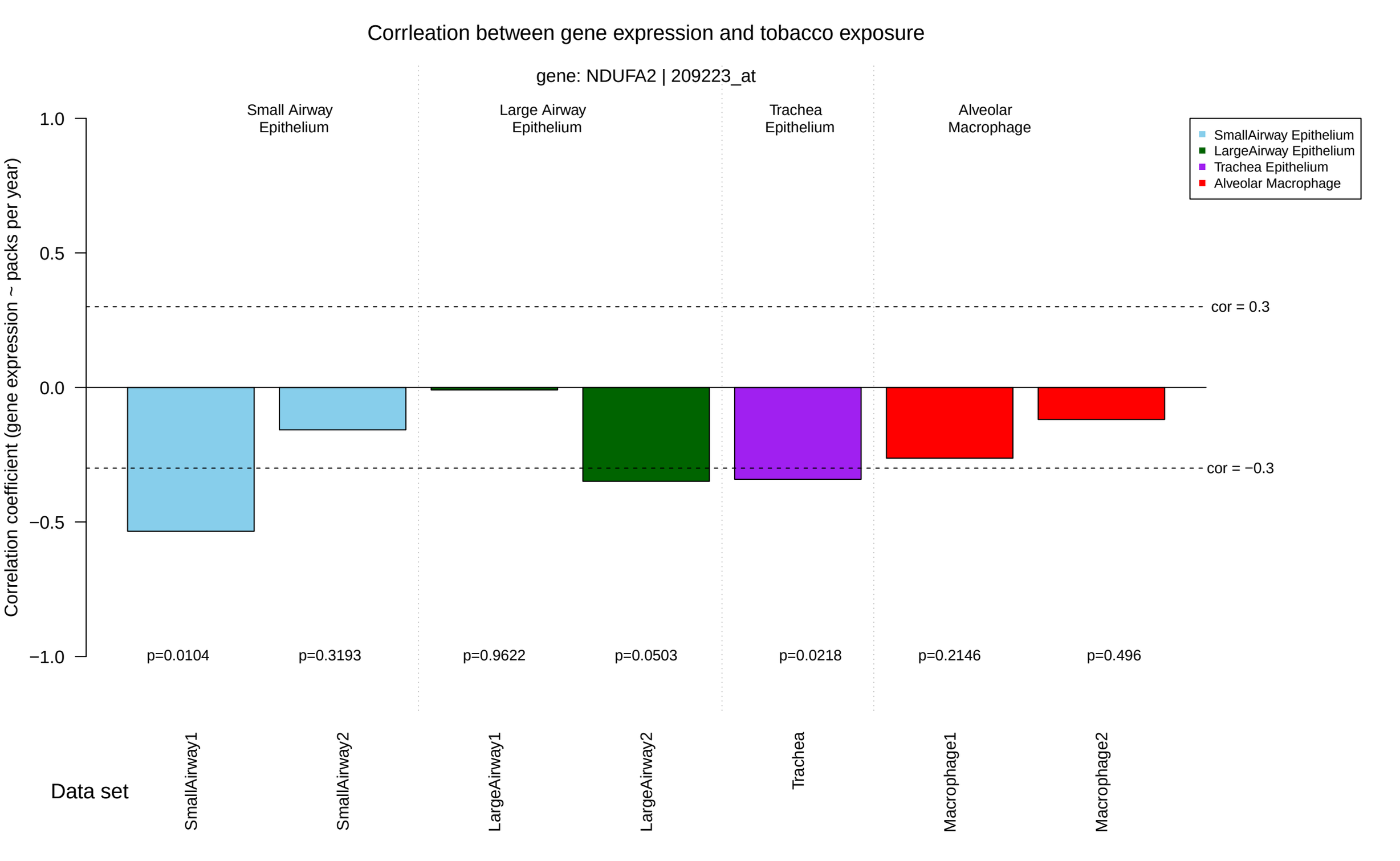

set: LargeAirway1 (p=0.4375)
NDUFA2 | 209223_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.161)
NDUFA2 | 209223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0111)
NDUFA2 | 209223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0617)
NDUFA2 | 209223_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0518)
NDUFA2 | 209223_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: NDUFA2 | 209223_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0104
p=0.3193
p=0.9622
p=0.0503
p=0.0218
p=0.2146
p=0.496
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2